** Autonomous Indoor Drone Navigation**

Installation Guide

This guide explains how to set up the complete simulation environment for the Autonomous Indoor Drone Navigation project using PX4 Autopilot, ROS 2 Humble, Gazebo Sim (Garden), and Micro XRCE-DDS Agent on Ubuntu 22.04 LTS.

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System Requirements

Component	Version
Operating System	Ubuntu 22.04.5 LTS (Jammy Jellyfish)
ROS Distribution	ROS 2 Humble Hawksbill
Flight Controller	PX4 Autopilot v1.14
Simulator	Gazebo Sim (Garden)

Communication Bridge	Micro XRCE-DDS Agent v2.4.2
Build System	Colcon + CMake

Software Stack

```
Gazebo Sim (Garden)
     |
Sensor Simulation
     |
     ▼
PX4 SITL (Flight Controller)
     |
XRCE-DDS (UDP Port 8888)
     |
     ▼
Micro XRCE-DDS Agent
     |
ROS 2 DDS Network
     |
     ▼
ROS 2 Humble Workspace
```

---

1. Install Ubuntu 22.04 LTS version

   Check Version

   ```
   lsb_release -a
   ```

   ```
   Distributor ID:	Ubuntu
   Description:	Ubuntu 22.04.5 LTS
   Release:	22.04
   Codename:	jammy
   ```

2. Update System

```
sudo apt update
sudo apt upgrade -y
sudo apt install git curl wget build-essential cmake python3-pip python3-colcon-common-extensions -y
```

3. Install the ROS2 distribution [source]

Check Version

```
echo $ROS_DISTRO
Humble
```

4. Install Colcon + ROS Tools

```
sudo apt install python3-colcon-common-extensions python3-rosdep python3-vcstool -y
sudo rosdep init
rosdep update
```

4. Install PX4 Autopilot v1.14 [source]

```
cd ~
git clone -b v1.14.0 --recursive https://github.com/PX4/PX4-Autopilot.git
cd ~/PX4-Autopilot
bash ./Tools/setup/ubuntu.sh --no-sim-tools
```

Remember to **reboot** your machine after this script runs for the first time.

Build PX4 (After reboot):

```
cd ~/PX4-Autopilot
make px4_sitl
```

Check Version

```
cd ~/PX4-Autopilot
git describe --tags

v1.14.0
```

5. Install gazebo [source]

```
sudo apt update && sudo apt install -y curl lsb-release gnupg
sudo wget https://packages.osrfoundation.org/gazebo.gpg -O /usr/share/keyrings/pkgs-osrf-archive-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/pkgs-osrf-archive-keyring.gpg] http://packages.osrfoundation.org/gazebo/ubuntu-stable $(lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/gazebo-stable.list > /dev/null

sudo apt update
sudo apt install gz-garden
sudo apt install ros-humble-ros-gzgarden


echo "export GZ_VERSION=garden" >> ~/.bashrc
source ~/.bashrc

sudo apt update && sudo apt install -y \
  ros-humble-desktop \
  ros-humble-ros-gzgarden-bridge \
  ros-humble-octomap \
  ros-humble-octomap-server \
  ros-humble-octomap-msgs \
  ros-humble-octomap-ros \
  ros-humble-octomap-rviz-plugins \
  ros-humble-pcl-ros
```

Check Version

```
gz sim --version
Gazebo Sim, version 7.9.0
Copyright (C) 2018 Open Source Robotics Foundation.
Released under the Apache 2.0 License.
```

6. Micro XRCE-DDS Agent Installation [source]

```
git clone -b v2.4.2 https://github.com/eProsima/Micro-XRCE-DDS-Agent.git
cd Micro-XRCE-DDS-Agent
mkdir build
cd build
cmake ..
make
sudo make install
sudo ldconfig /usr/local/lib/
```

Check Version

```
cd ~/Micro-XRCE-DDS-Agent
git describe --tags
```

```
v2.4.2
```

7. Install PX4 ROS2 Messages

```
mkdir -p ~/px4_ros2_ws/src
cd ~/px4_ros2_ws/src
git clone https://github.com/PX4/px4_msgs.git -b release/1.14
cd ~/px4_ros2_ws
source /opt/ros/humble/setup.bash
colcon build
```

- After building, source before running any code

```
source ~/px4_ros2_ws/install/setup.bash
```

- Otherwise

```
echo "source ~/px4_ros2_ws/install/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

8. If you want to rebuild colcon after some changes in file

```
cd ~/px4_ros2_ws
colcon build --symlink-install
source install/setup.bash
```

9. Kill all the background command

```
killall -9 px4
killall -9 ruby
killall -9 MicroXRCEAgent
pkill -9 -f "gz sim"
pkill -9 -f "parameter_bridge"
```